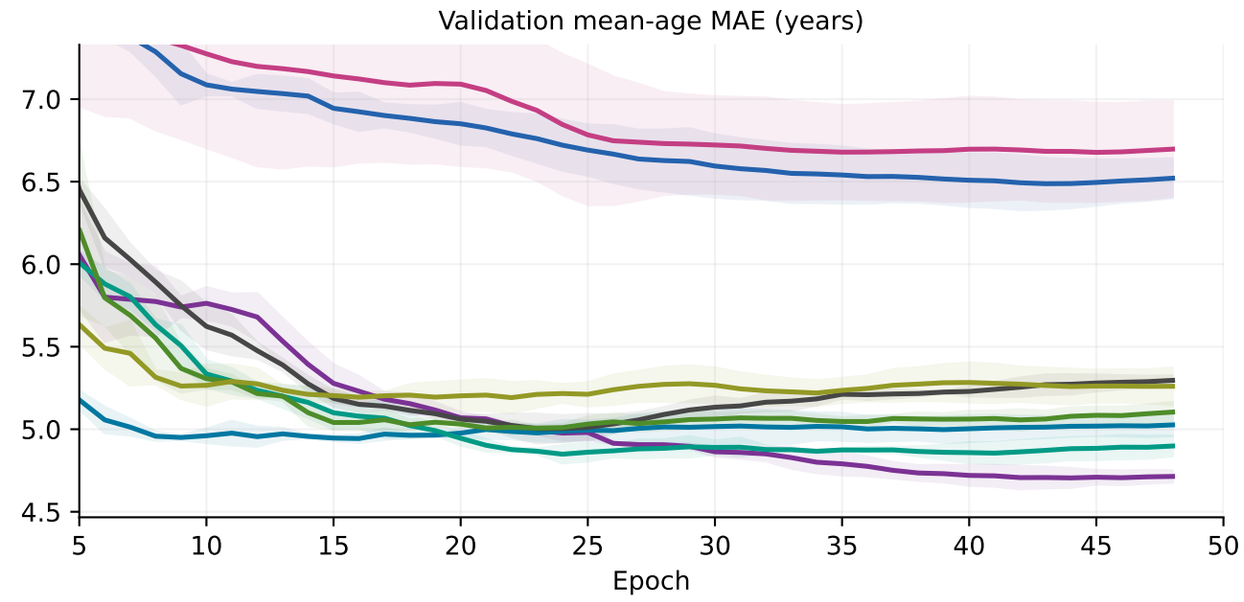
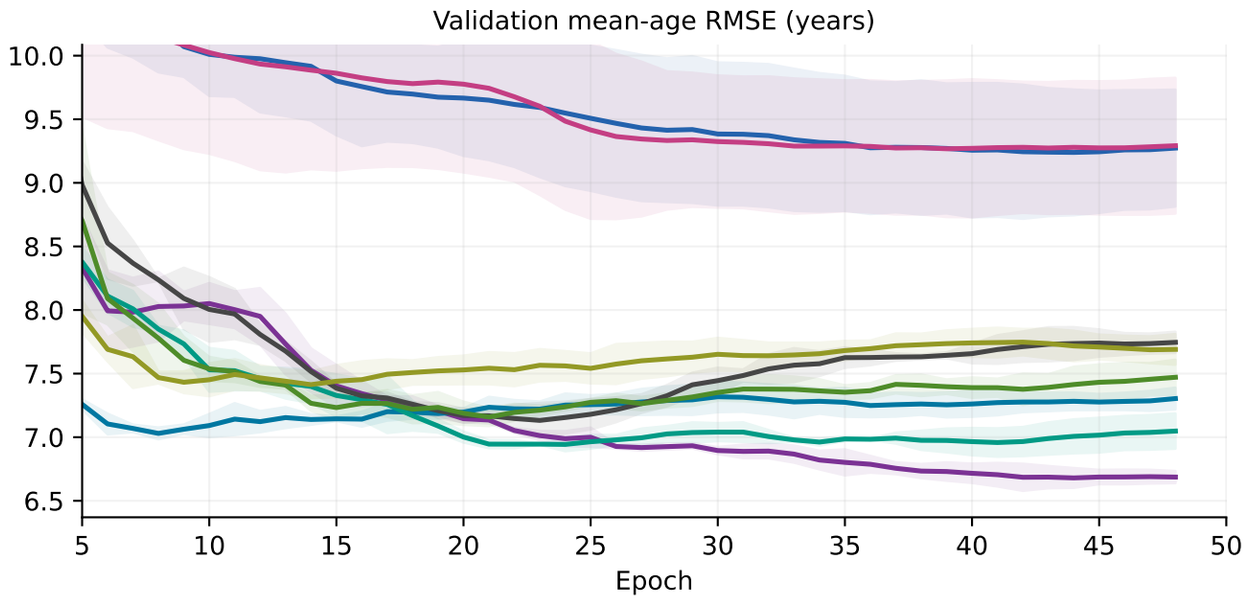
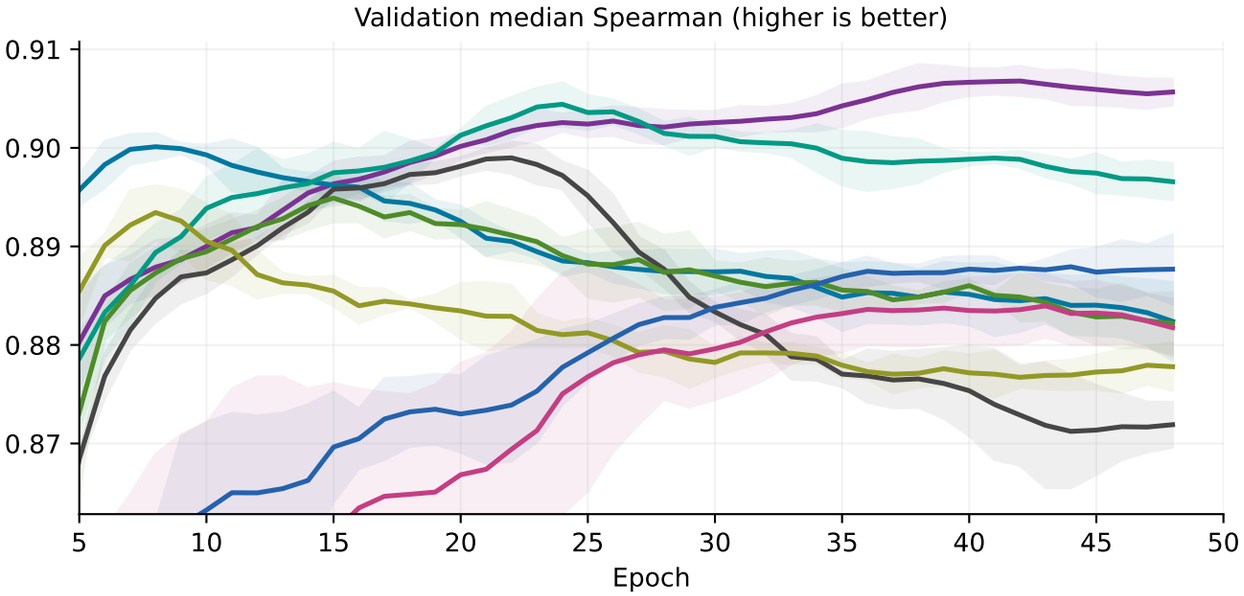
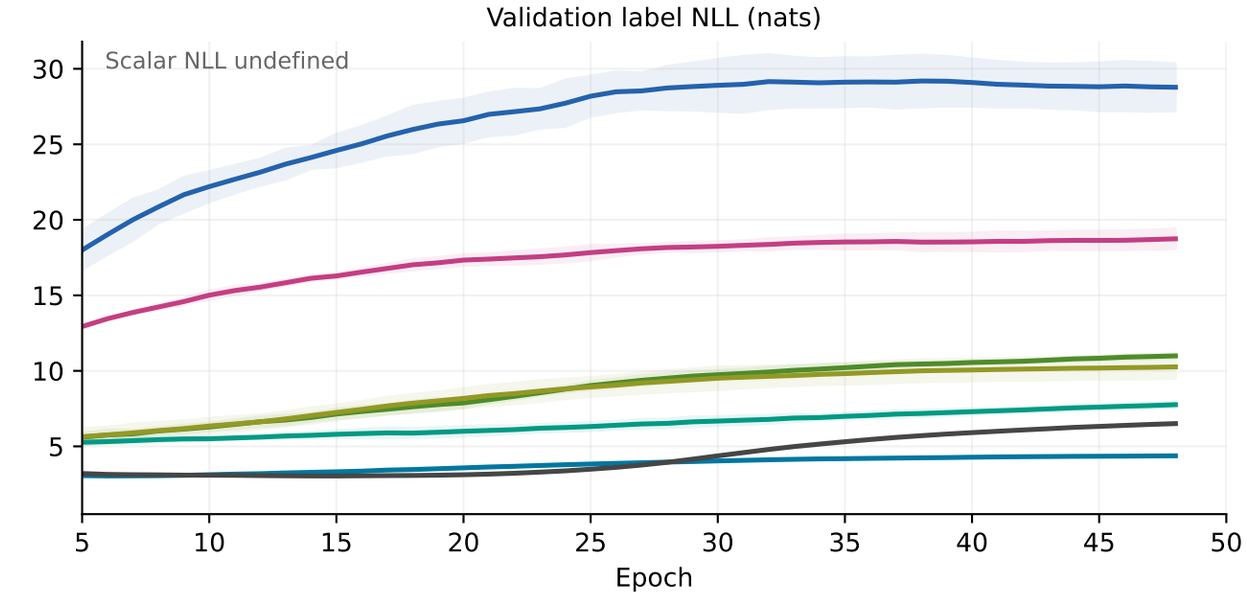
Full data: eight-method comparison







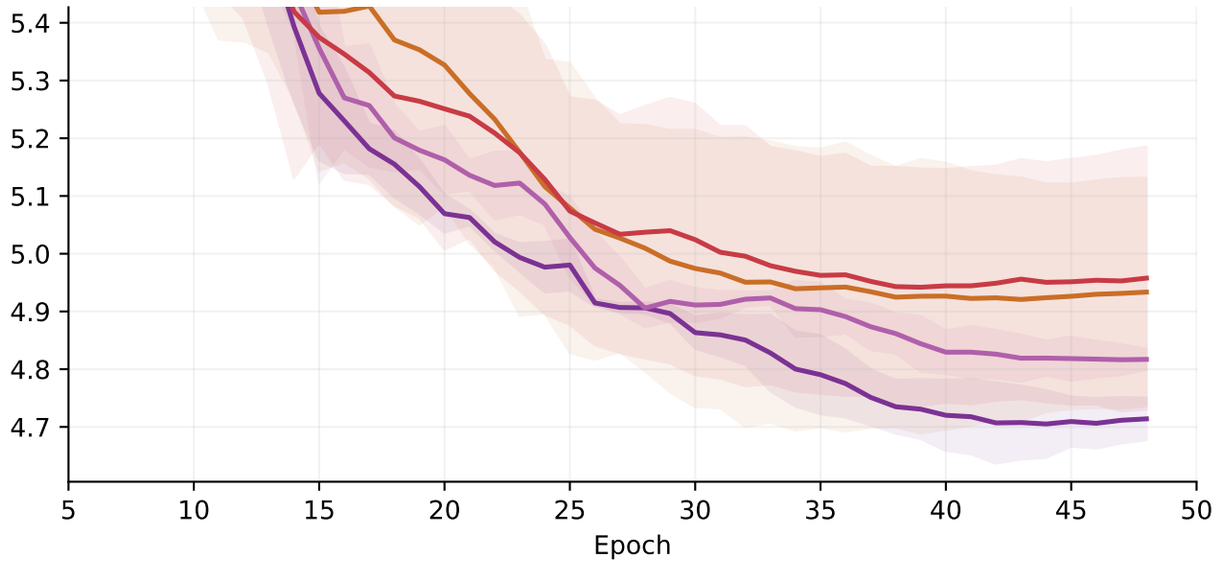


Scalar MAE
TailRL (population)
TailRL (64 samples)
CE
Stat MAE, w=2 (population)
Stat MSE, w=2 (population)
GRPO MAE (64 samples)
RLOO MAE (population)

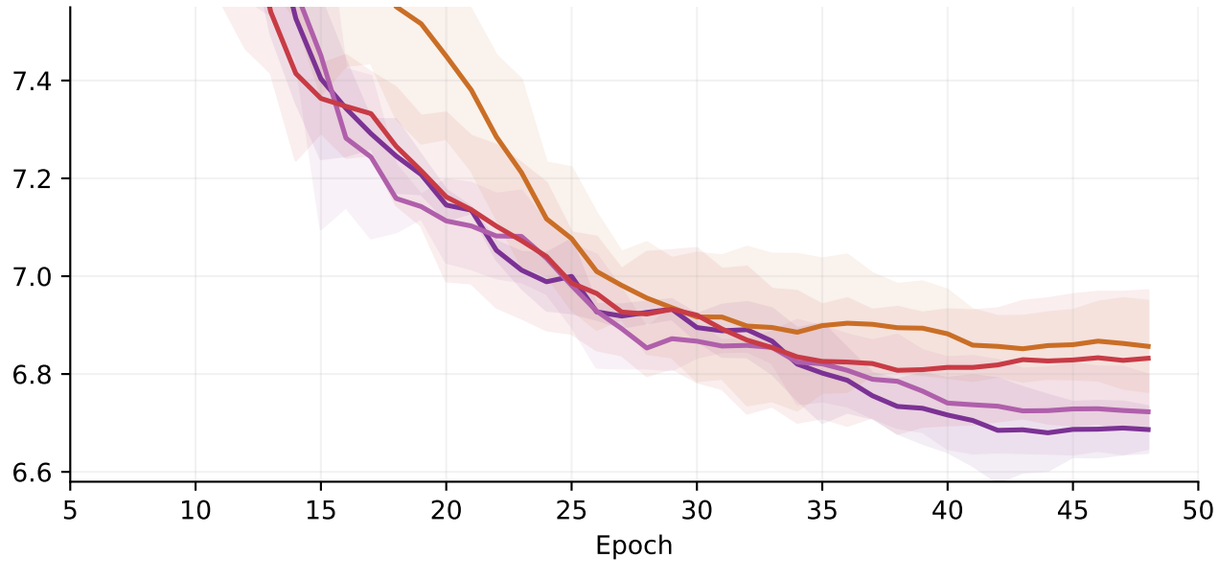
18,462 images × 50 epochs; three matched seeds. GRPO/RLOO configurations were selected by mean best validation MAE, without test outcomes. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.

Full data: scalar and categorical SL

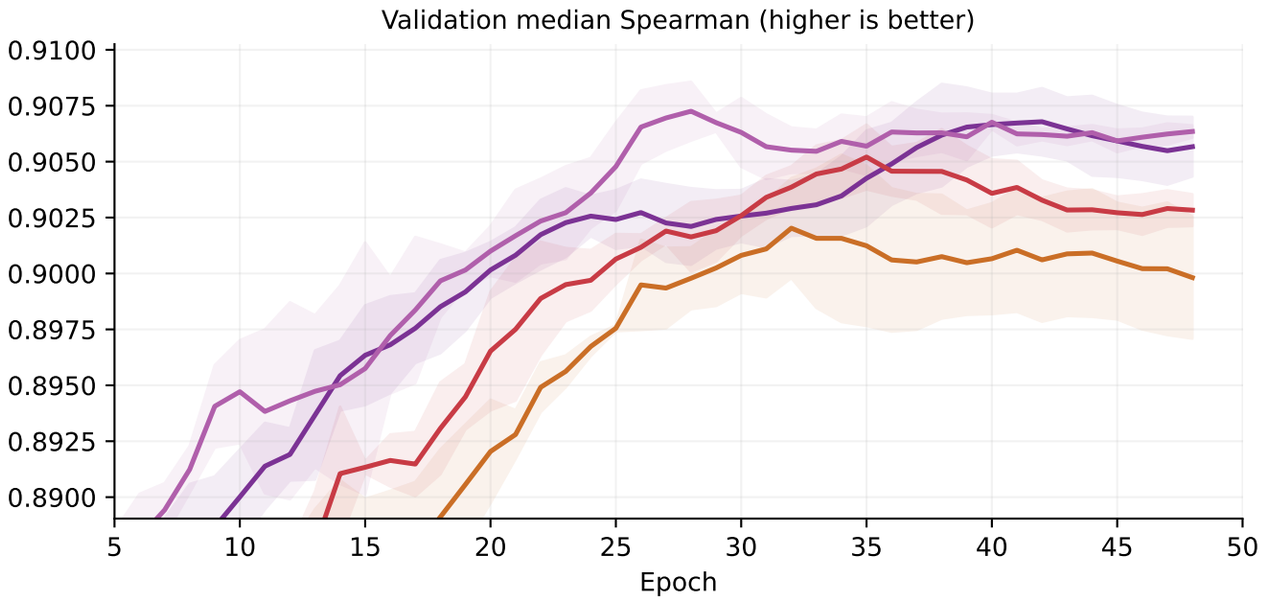
Validation mean-age MAE (years)



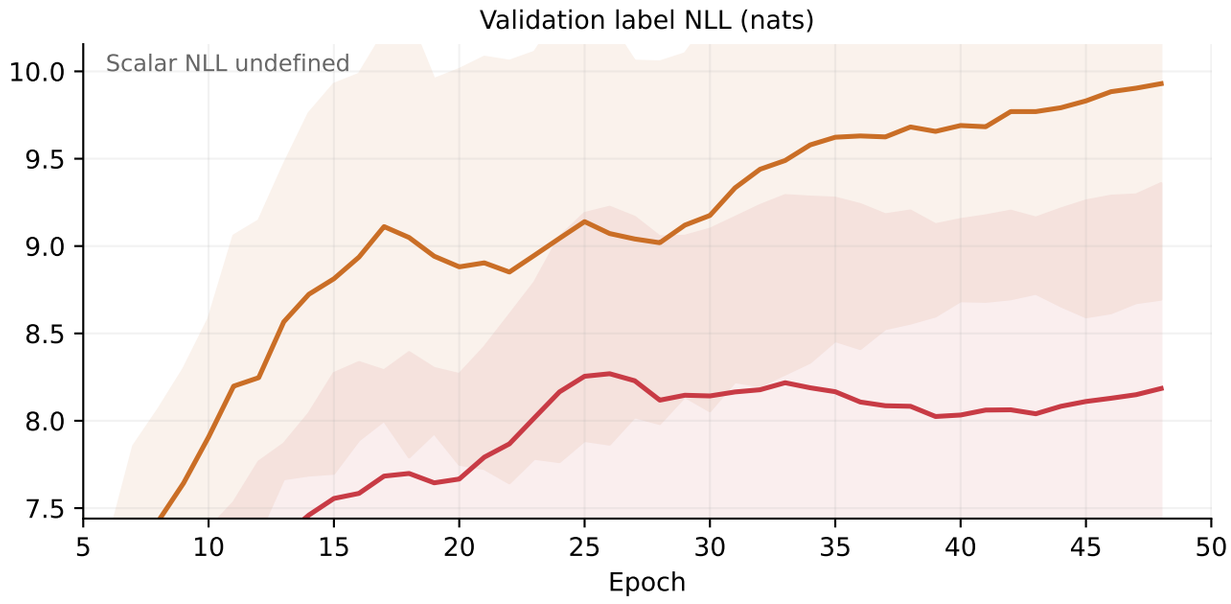
Validation mean-age RMSE (years)

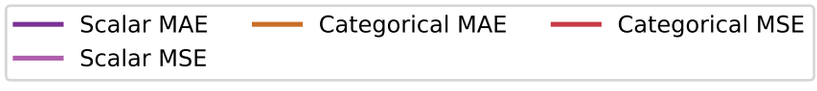


Validation median Spearman (higher is better)



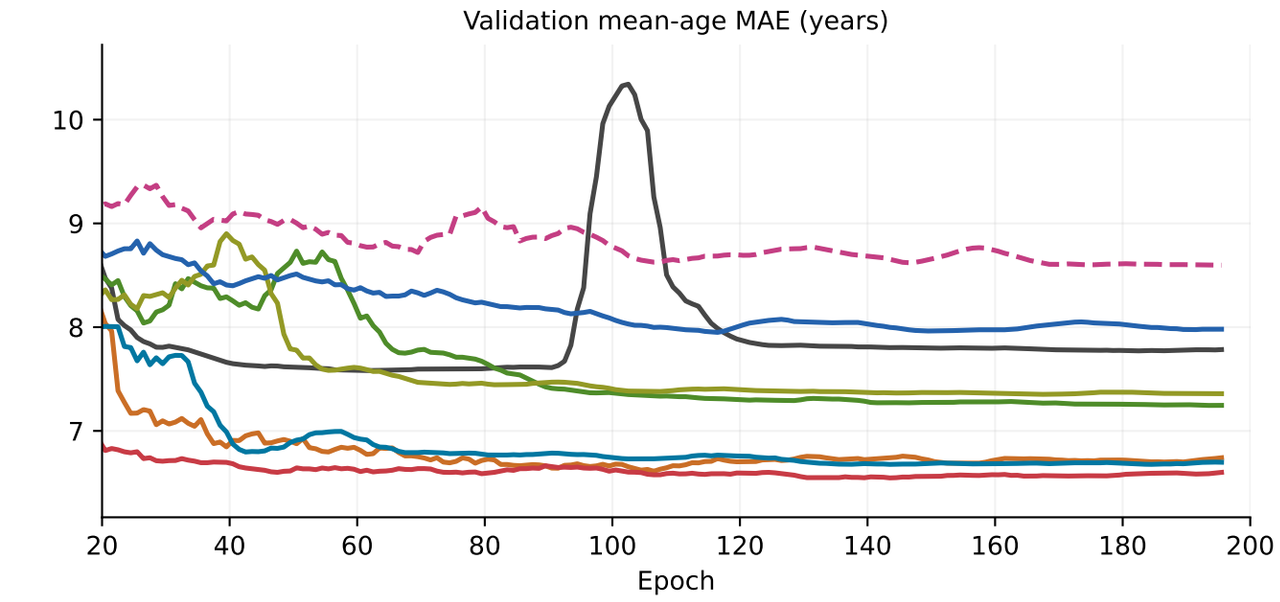
Validation label NLL (nats)

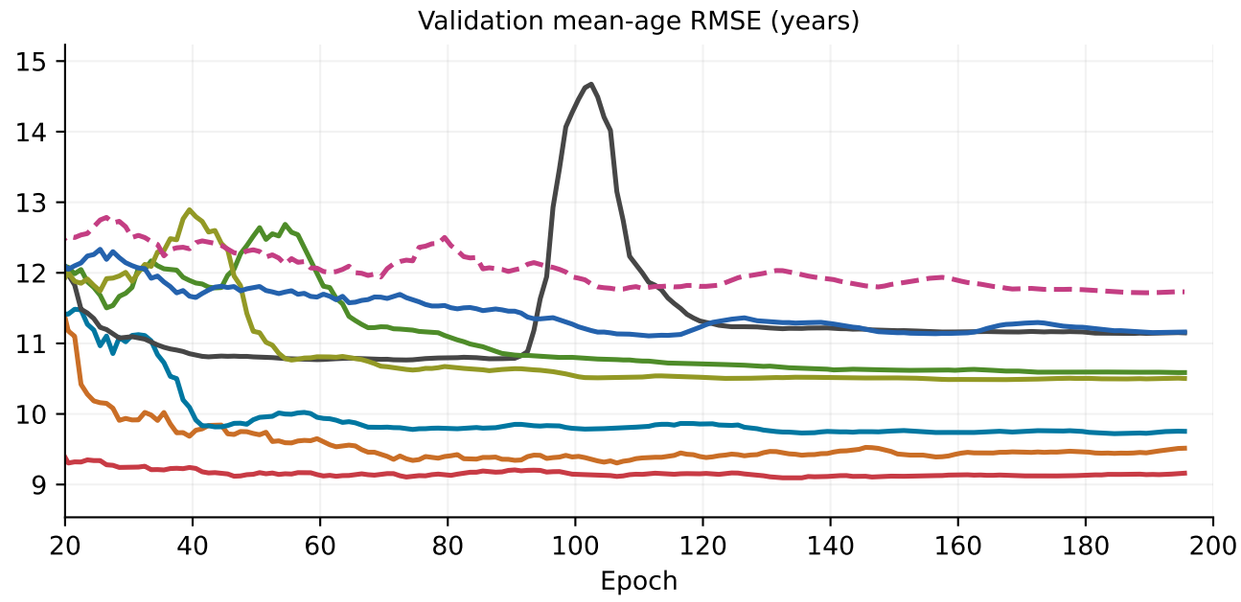


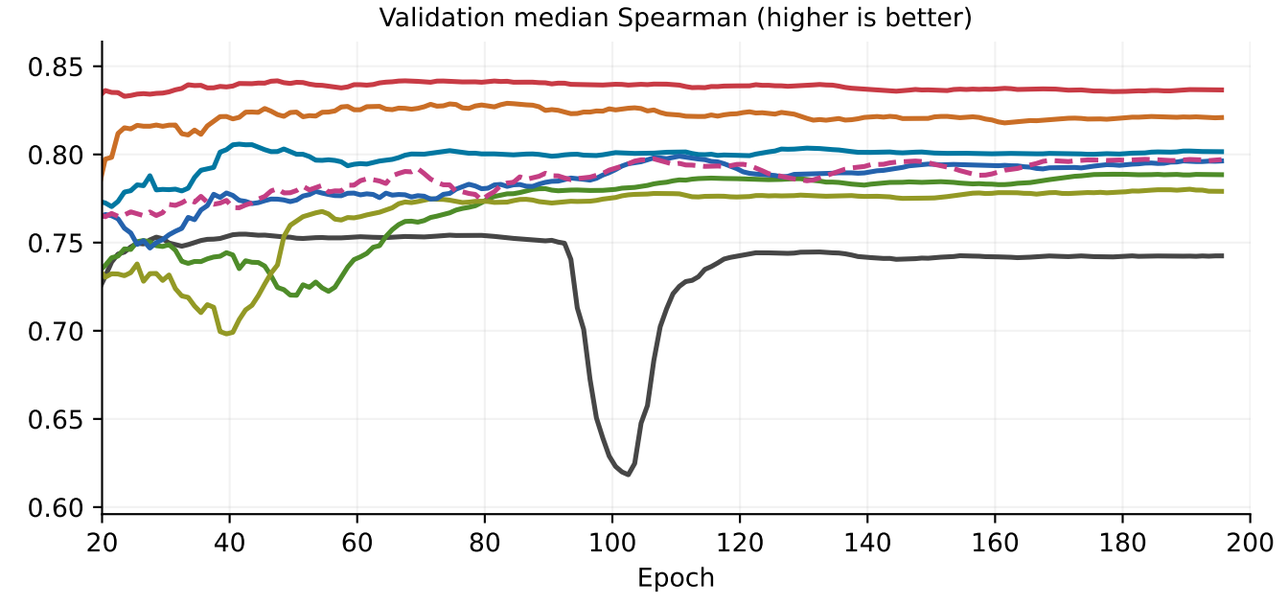


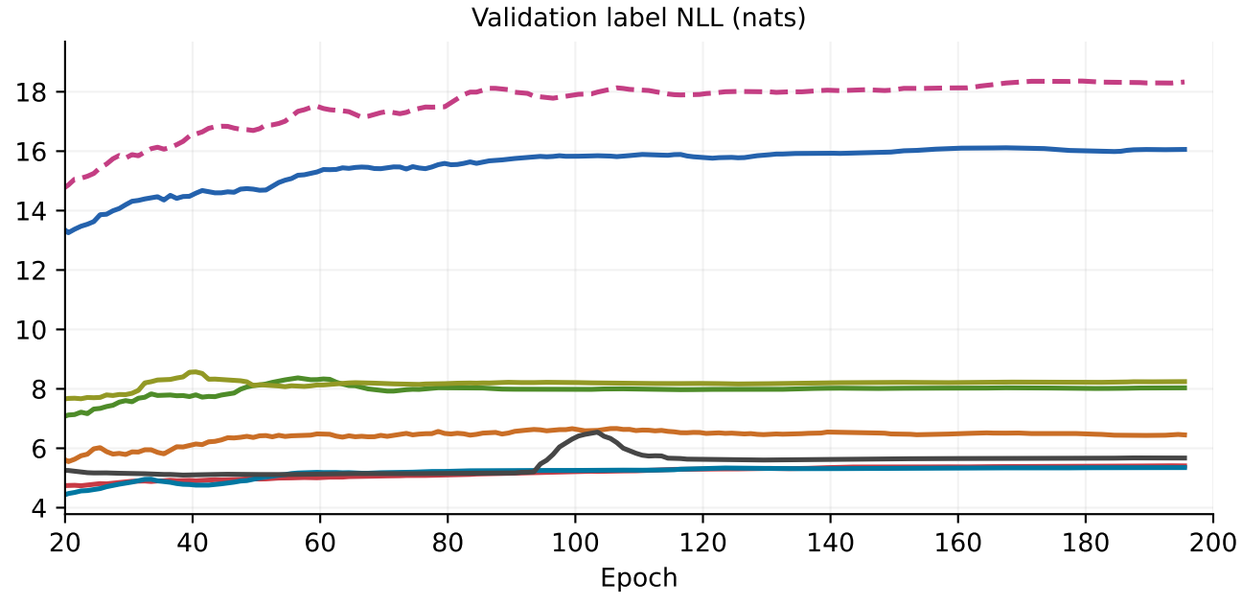
Three matched seeds; all four settings use peak LR 0.001. Scalar label NLL is undefined and is omitted from that panel. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.

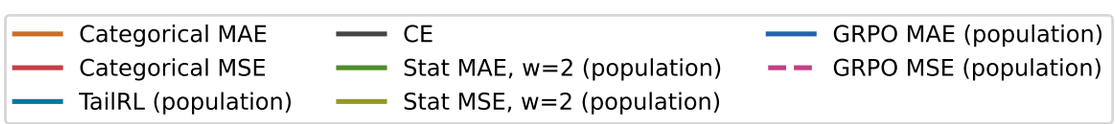
1024 images × 200 epochs: eight tested methods



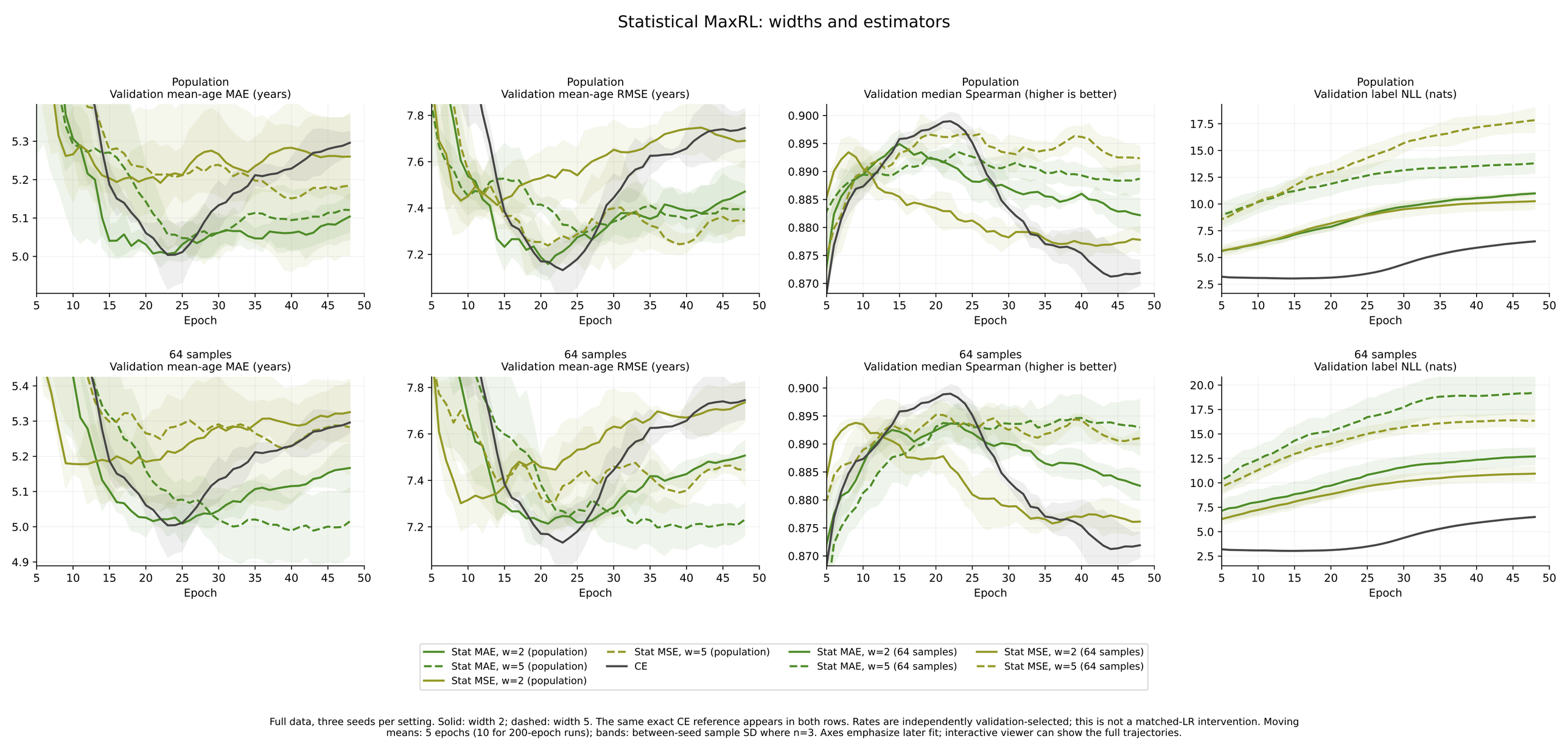


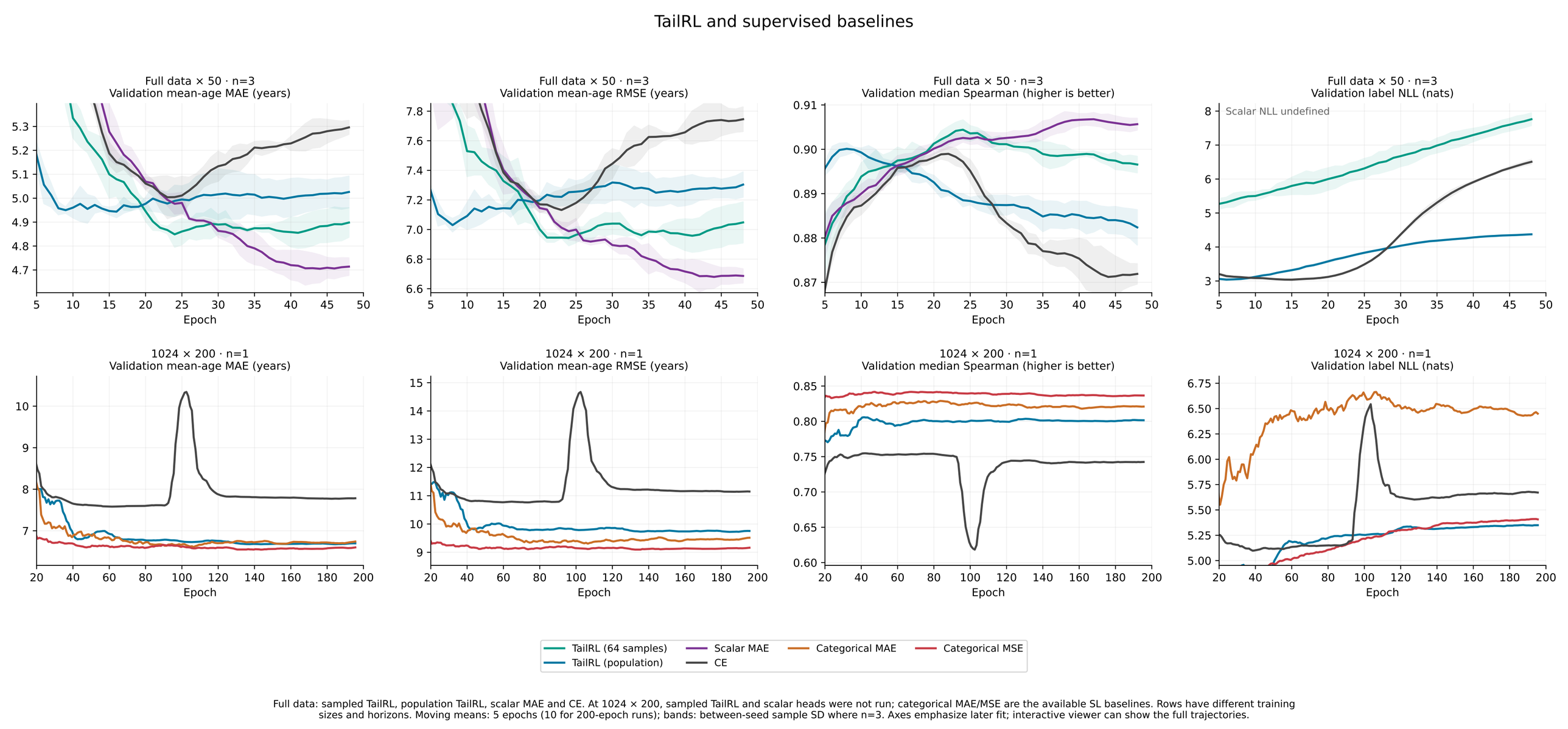




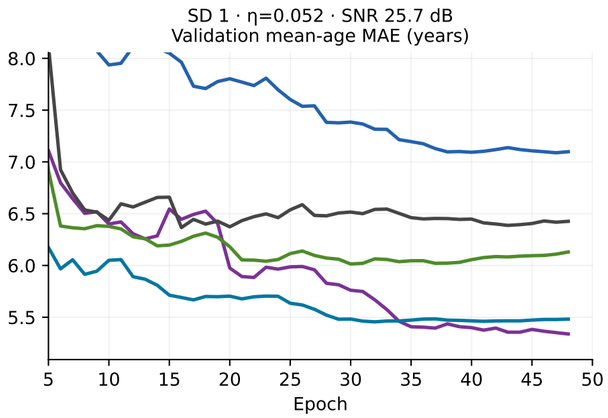


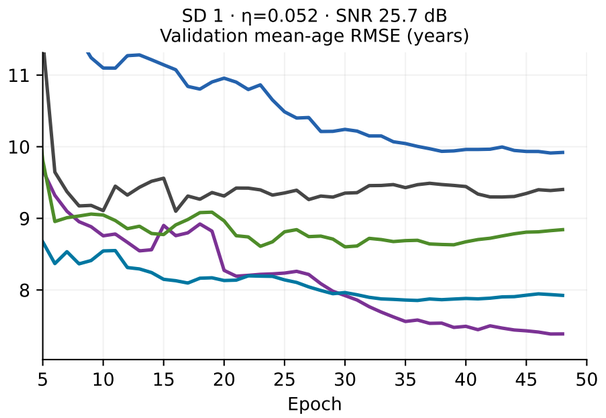
Seed 42. Adapted to the eight completed categorical settings: categorical MAE/MSE and both population GRPO rewards replace the unrun scalar, sampled-TailRL and RLOO settings. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.

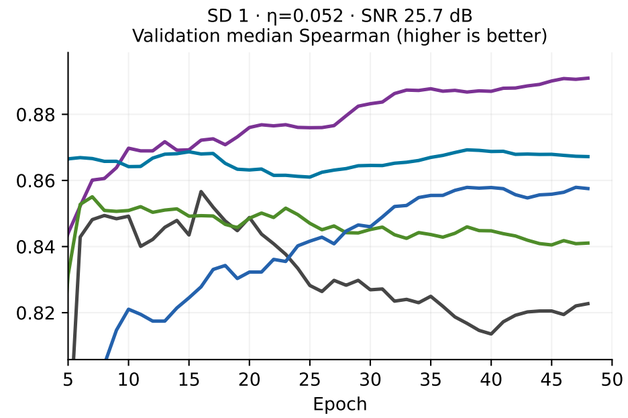

Statistical MaxRL: widths and estimators
Population
Validation mean-age MAE (years)
Population
Validation mean-age RMSE (years)
Population
Validation median Spearman (higher is better)
Population
Validation label NLL (nats)
64 samples
Validation mean-age MAE (years)
64 samples
Validation mean-age RMSE (years)
64 samples
Validation median Spearman (higher is better)
64 samples
Validation label NLL (nats)
Epoch
Stat MAE, w=2 (population)
Stat MAE, w=5 (population)
Stat MSE, w=2 (population)
Stat MSE, w=5 (population)
CE
Stat MAE, w=2 (64 samples)
Stat MAE, w=5 (64 samples)
Stat MSE, w=2 (64 samples)
Stat MSE, w=5 (64 samples)
Full data, three seeds per setting. Solid: width 2; dashed: width 5. The same exact CE reference appears in both rows. Rates are independently validation-selected; this is not a matched-LR intervention. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.

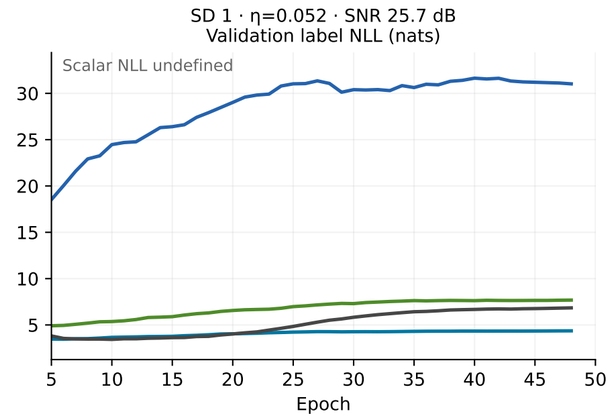

TailRL and supervised baselines
Full data × 50 · n=3
Validation mean-age MAE (years)
Full data × 50 · n=3
Validation mean-age RMSE (years)
Full data × 50 · n=3
Validation median Spearman (higher is better)
Full data × 50 · n=3
Validation label NLL (nats)
Scalar NLL undefined
1024 × 200 · n=1
Validation mean-age MAE (years)
1024 × 200 · n=1
Validation mean-age RMSE (years)
1024 × 200 · n=1
Validation median Spearman (higher is better)
1024 × 200 · n=1
Validation label NLL (nats)
Epoch
TailRL (64 samples)
TailRL (population)
Scalar MAE
CE
Categorical MAE
Categorical MSE
Full data: sampled TailRL, population TailRL, scalar MAE and CE. At 1024 × 200, sampled TailRL and scalar heads were not run; categorical MAE/MSE are the available SL baselines. Rows have different training sizes and horizons. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.

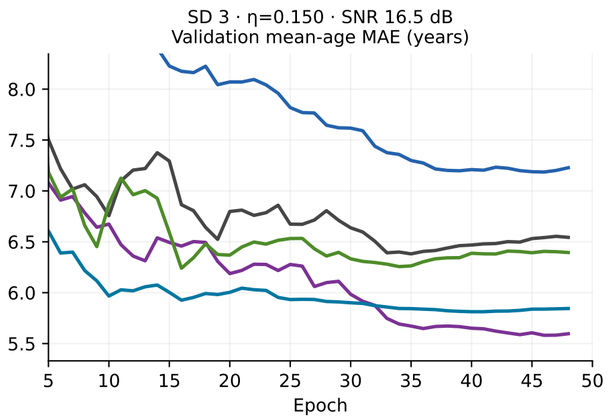
Fixed training-label noise: four levels

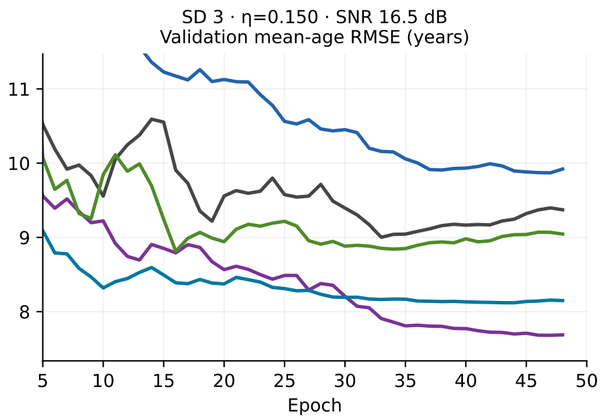


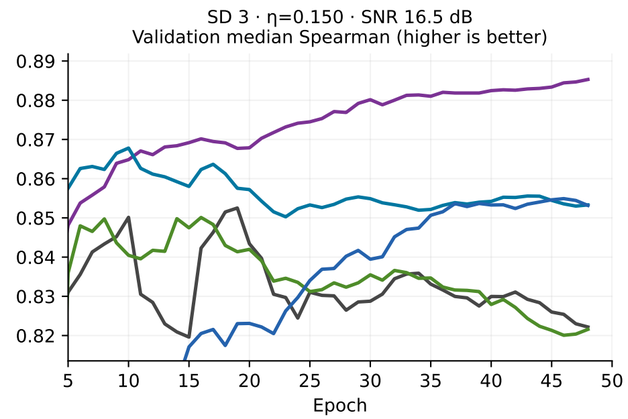


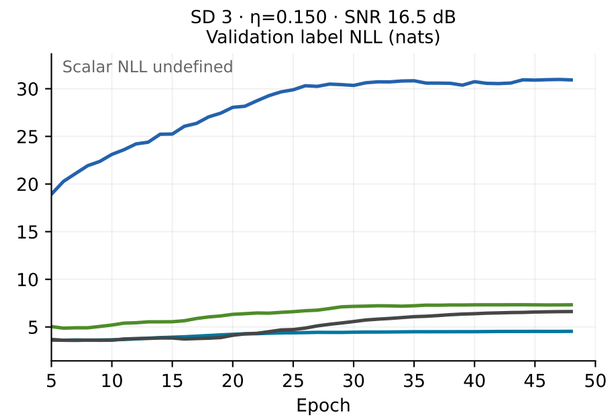


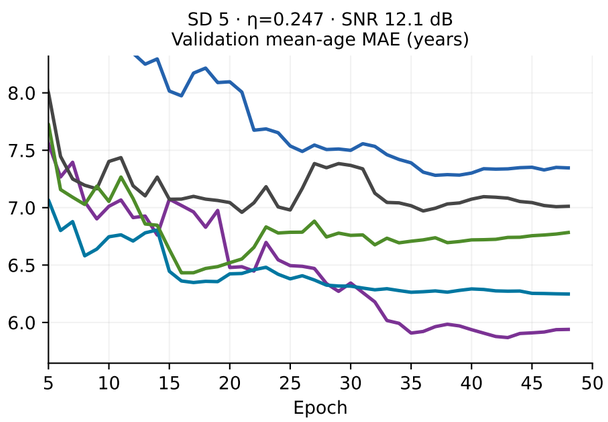


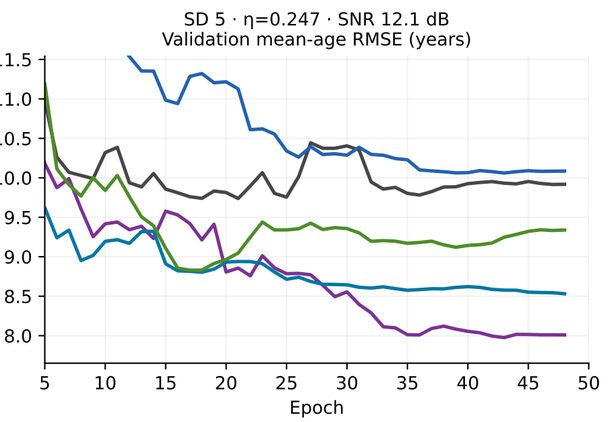


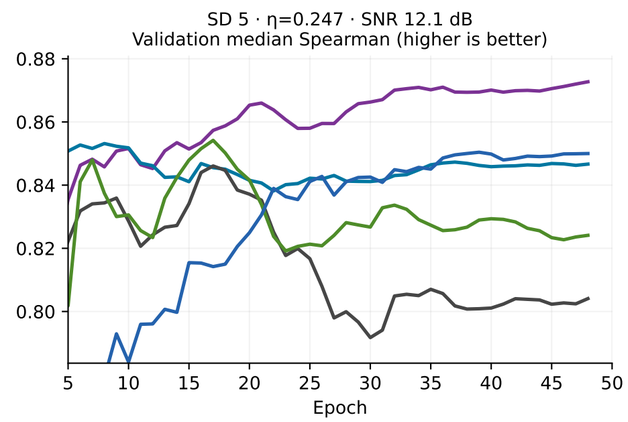


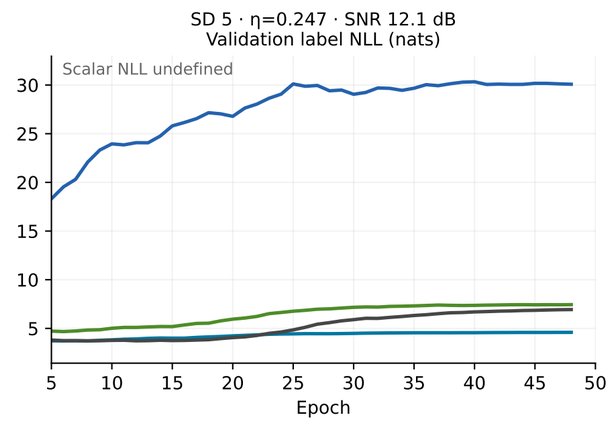


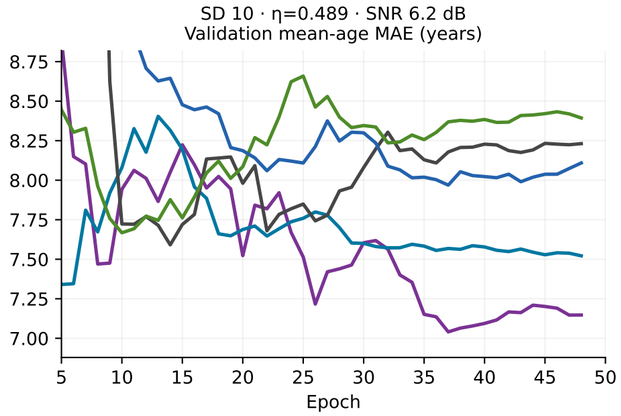


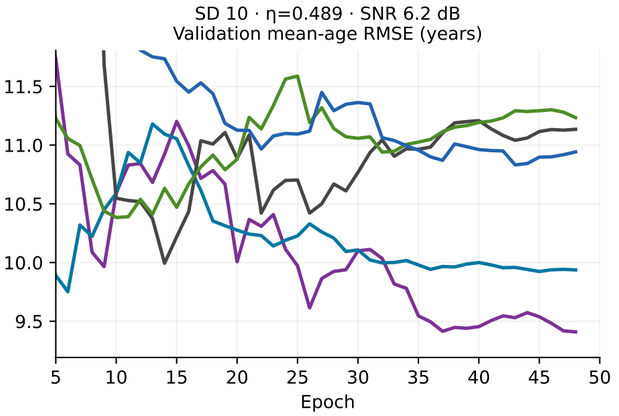


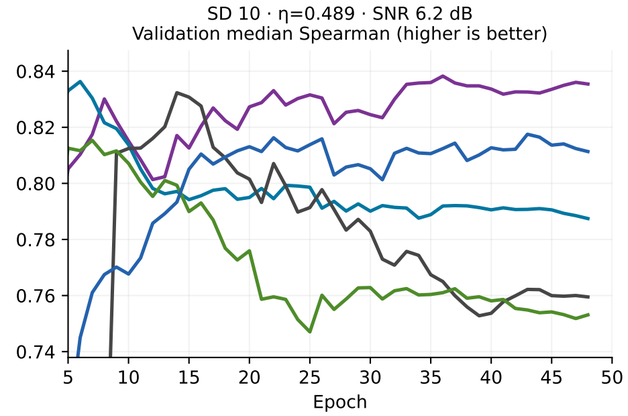


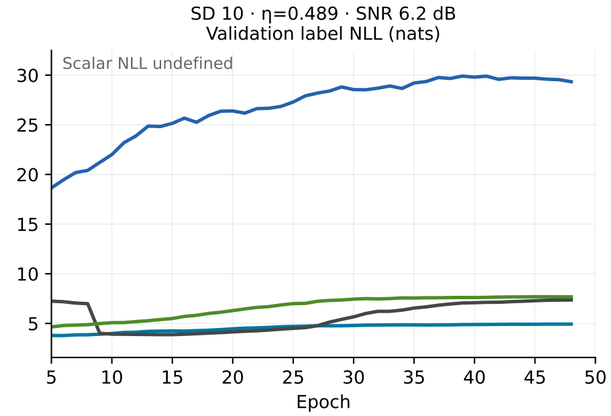




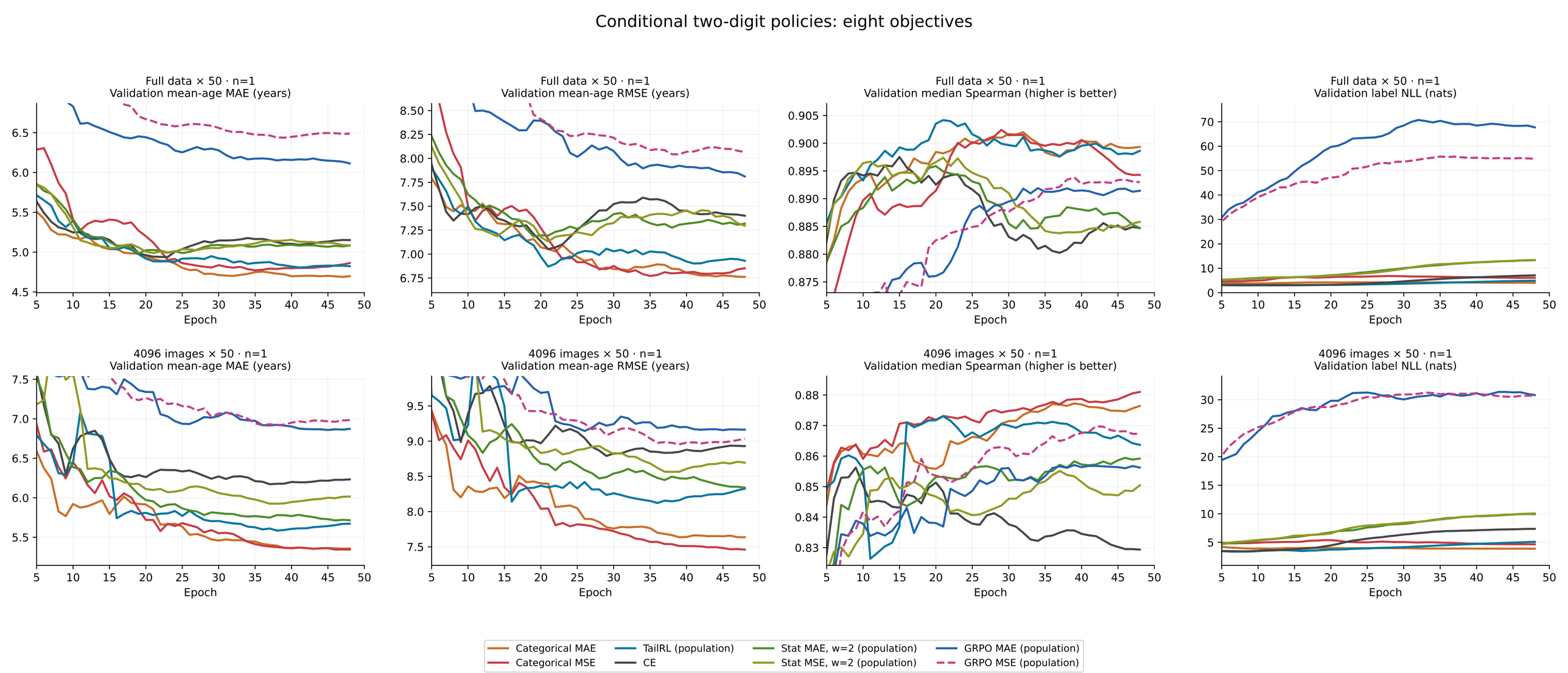








4096 images × 50 epochs; optimization seed 42, noise seed 2026; held-out labels are clean. η = noise RMS / clean-age SD (about 0.05 at SD 1); conventional power SNR = 1/η². SD 3 replaces the unrun SD 2 by request. GRPO MAE is the best available clean-validation RL configuration; RLOO noise arms were not run. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.



Seed 42. Eight conditional 10×10 objectives on full data and 4096 images. All digit rates were selected on 4096 images; full-data digit runs reuse them. These panels contain digit policies only. Moving means: 5 epochs (10 for 200-epoch runs); bands: between-seed sample SD where n=3. Axes emphasize later fit; interactive viewer can show the full trajectories.